

FACULTY OF SCIENCE, ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CSE 113: Discrete Mathematics

Batch: 27th

Semester: Fall 2024

Date: 22/03/2025

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings.

5x8= 40

1. (a) Define Hasse Diagram.

CLO 4 2

(b) Let $A = \{1, 2, 3, 24, 48, 96, 192\}$ is a set. Find out $R = \{(a, b) \mid a \text{ divides } b\}$. Draw the partial ordering diagram for R and finalize the Hasse diagram from the partial ordering diagram, also find out the maximal and minimal.

CLO 4 6

2. (a) Write the pseudocode for Kruskal's algorithm to determine Minimum Spanning Tree (MST).

CLO 2 3

(b) Let us consider the following 2 matrices.

CLO 4 5

$$A = \begin{pmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 1 & -2 \end{pmatrix} \quad C = \begin{pmatrix} -2 & 3 & 9 \\ 8 & -11 & -34 \\ -5 & 7 & 21 \end{pmatrix}$$

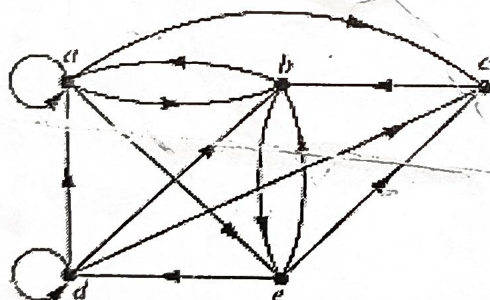
Find $A+C$, $A \times C$ and the inverse of the given matrix A ,

3. (a) Suppose that a connected planar simple graph has 30 vertices, each of degree 3. How many regions do a representation of this planar graph split the plane?

CLO 4 2

(b)

CLO 4 6x1
= 6



Find the followings of the above di-graph:

- i) properties with reason,
- ii) adjacency matrix,
- iii) adjacency list,
- iv) degree,
- v) in-degree and
- vi) out-degree

4. (a) Find the general equation for the Pizza-cut problem.

CLO 4 2

(b) Briefly explain the Tower of Hanoi Problem.

CLO 4 2

(c) Let $A = \{1, 2, 4, 12, 24, 48\}$ is a set, Determine the following relations:

CLO 4 4

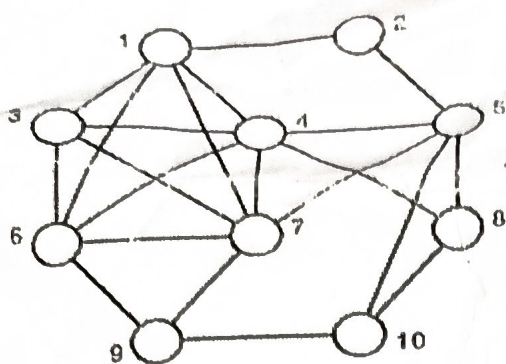
i) $R_1 = \{(a, b) \mid a > b\}$ and

ii) $R_2 = \{(a, b) \mid a^2 + b \leq 200\}$

Find the properties with reason of each relation and represent them in matrix and graphically.

5. (a) Color the following graph by using the Four color theorem:

CLO 4 2



- (b) Prove that an undirected graph has an even number of vertices of odd degree.

CLO 4 3

- (c) Briefly explain the Handshaking theorem of a graph with example.

CLO 4 3

6. (a) Write the similarities and dissimilarities among the adjacency matrix, the adjacency list and the adjacency multilist of a graph.

CLO 4 4

- (b) Write the characteristics of a tree.

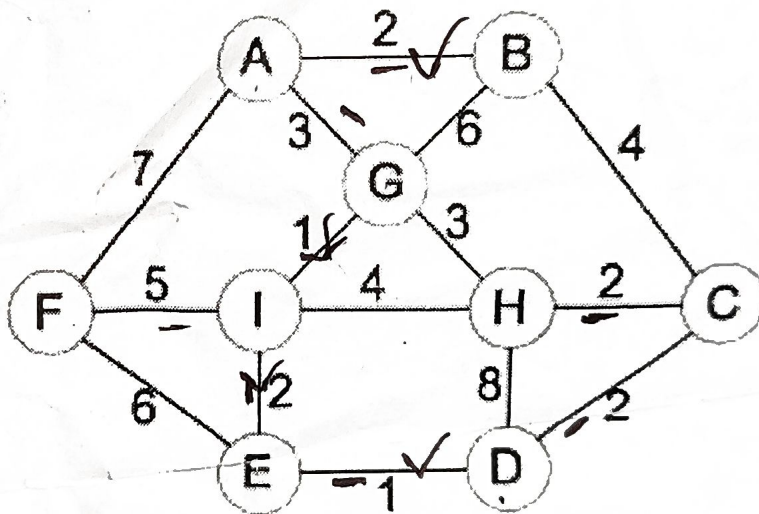
CLO 4 4

7. (a) Derive the MST for the following weighted graph by using

CLO 4 4

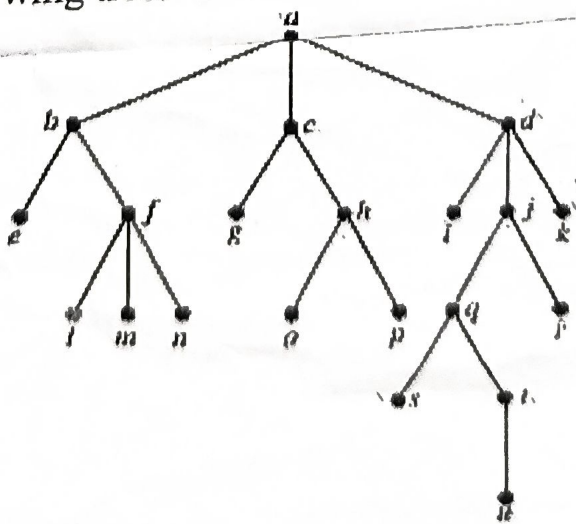
(i) Prim's algorithm

(ii) Kruskal's algorithm



- (b) Briefly describe tree traversal techniques and find the order of the elements after traversing the following tree:

CLO 4



Pre-Root-Left-Right
In-Left-Root-Right

Hamdard University Bangladesh

Dept. of Electrical and Electronic Engineering (EEE)

Final Examination (Fall'24)

EEE 117 Electrical Engineering

Time: 2 Hours

Marks: 40

Answer any 4 (Four) from the following sets of questions

1. (a) Write a short note on following questions:

- Thevenin's Theorem
- Norton's Theorem
- Maximum Power Transfer Theorem

3 CLO1

(b) Find the value of R_L for maximum power transfer in the circuit of Fig-1(b). Find the maximum power.

7 CLO3

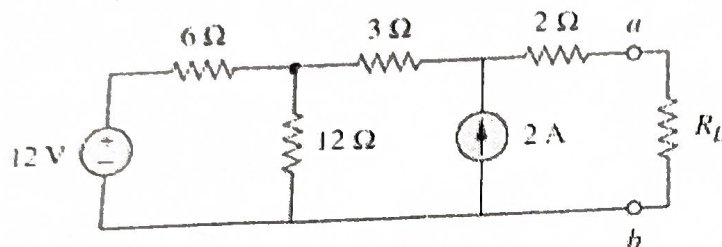


Fig-1(b)

2. (a) Consider the circuit in Fig-2(a). Under dc conditions, find: (a) i , and (b) the energy stored in the capacitor and inductor.

5 CLO3

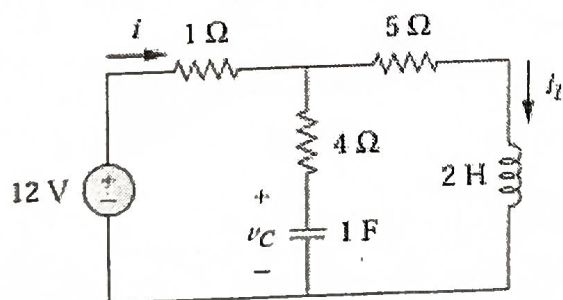


Fig-2(a)

(b) Determine the current through a $200 \mu\text{F}$ capacitor whose voltage is shown in Fig-2(b).

5 CLO3

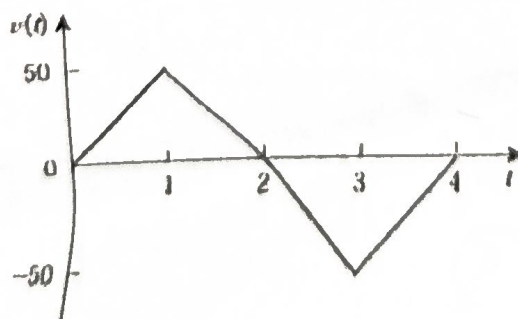


Fig-2(b)

3. (a) Derive the voltage response of a source free RC Circuit. Sketch the voltage decay curve and explain how the capacitor voltage behaves over time.

4 CLO2

- (b) In the circuit shown in Fig-3(b), find i_0 , v_0 , i for all time, assuming that the switch was open for a long time.

6 CLO3

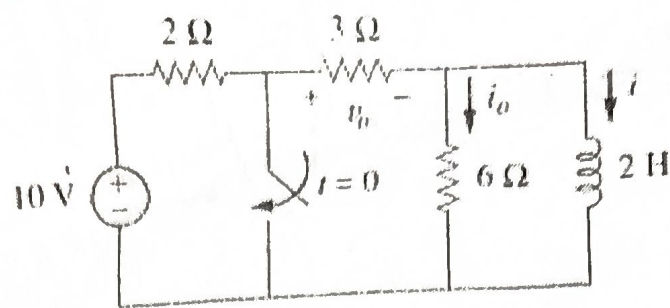


Fig-3(b)

- (a) Find current I in the circuit of Fig-4(a).

6 CLO3

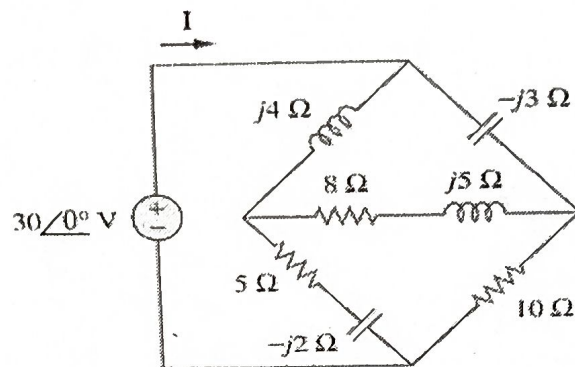


Fig-4(a)

- (b) Calculate v_0 in the circuit of Fig-4(b).

4 CLO2

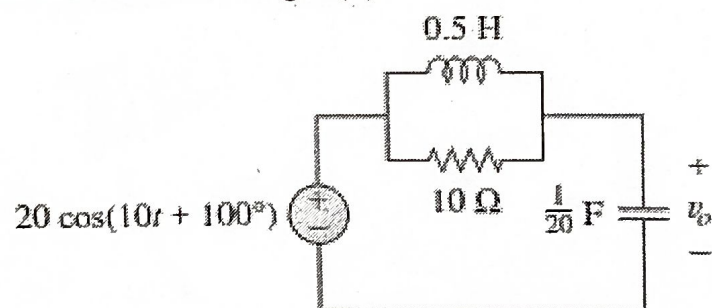


Fig-4(b)

- (a) Use the superposition theorem to find I_0 in the circuit in Fig-5(a).

6 CLO2

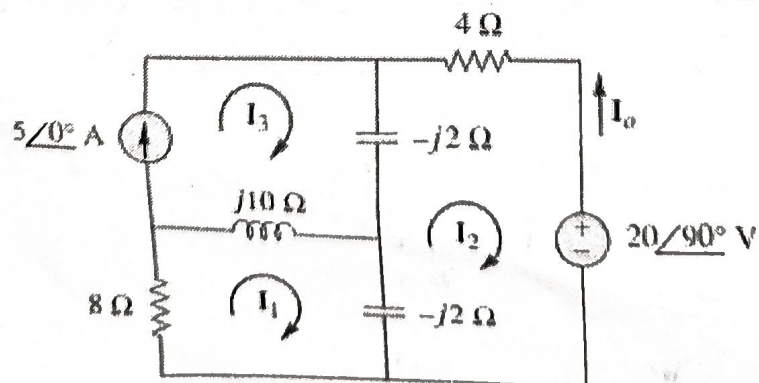


Fig-5(a)

(c) Obtain current in Fig-5(b) using Norton's theorem.

4 CLO2

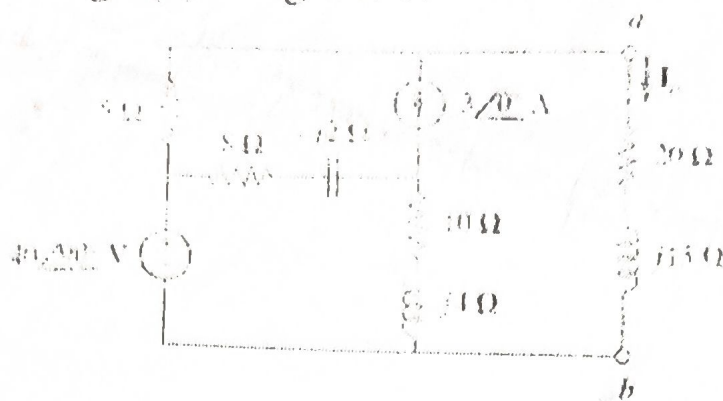


Fig-5(b)

6. (a) Derive the relation between magnetic flux, magnetomotive force (mmf) and reluctance in a magnetic circuit. 3 CLO2
- (b) Apply the analogy between electric and magnetic circuits to complete a table with related terms and symbols. 3 CLO2
- (c) Find phase angle between $i_1 = -4 \sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$. 4 CLO3

Does i_1 lead or lag i_2 ? Also find amplitude, phase, angular frequency, period and frequency for i_1 .

$$\omega = \frac{2\pi}{T}$$

$$\cos(\omega t + 90^\circ) = \sin \omega t$$

$$\sin(\omega t + 90^\circ) = \cos \omega t$$

$$\cos \omega t$$

$$-\cos(\omega t - 180^\circ) = \cos \omega t$$

$$-\sin(\omega t + 180^\circ) = \sin \omega t$$



STANDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

Course Code: MAT115

Course Title: Mathematics-I (Differential and Integral Calculus)

Final Examination

Semester: Fall 2024

Date: 3/17/2025

Total Time: 2 Hours

Total Marks: 40

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

	CLOs	Marks
Q1. (a) Define indeterminate forms.	CLO1	[1]
(b) State and prove the L-Hospital theorem.	CLO2	[4]
(c) Determine the following (Any two):	CLO3	[5]
(i) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5}$		
(ii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2 \cos x}{x \sin x}$		
(iii) $\lim_{x \rightarrow 0} \frac{x - \sin x \cos x}{x^3}$		
Q2. (a) State Rolle's theorem.	CLO2	[2]
(b) Verify Rolle's theorem for the function $f(x) = \frac{1}{2-x^2}$ in the interval $[-1, 1]$.	CLO3	[4]
(c) Discuss the applicability of mean value theorem for the function $f(x) = x(x-1)(x-3)$ in the interval $[0, 4]$.	CLO3	[4]
Q3. (a) Define the first and second Eulerian integrals.	CLO1	[2]
(b) State and prove the Taylor's theorem.	CLO2	[4]
(c) Evaluate the following integrals (any two):	CLO3	[4]
(i) $\int x^2 (\ln x)^2 dx$		
(ii) $\int \sin 3x \sin 7x dx$		
(iii) $\int_0^1 \frac{x^3 \tan^{-1} x^4}{1+x^8} dx$		
Q4. (a) Evaluate the following definite integrals (any one):		
(i) $\int_0^{\frac{\pi}{2}} \sin^3 x \cos 2x dx$		
(ii) $\int_0^{\frac{\pi}{2}} \cos^3 x \sin 4x dx$	CLO3	[3]



HAMDARD UNIVERSITY BANGLADESH

- (b) Obtain the reduction formula for $I_n = \int e^{ax} \cos^n x \, dx$ and hence determine I_5 with $a = -4$. CLO2, [7]
CLO3
- Q5. (a) State and prove the Walli's theorem and hence deduce $\int_0^{\frac{\pi}{2}} \cos^{12} x \, dx$. CLO2, [5]
CLO3
- (b) Prove that, CLO2 [5]
$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$
- Q6. (a) Obtain the reduction formula for $I_n = \int \sec^n x \, dx$ and hence determine I_7 . CLO2, [4]
CLO3
- (b) Determine the following integrals in terms of partial fractions: CLO3 [6]
(i) $\int \frac{x^2}{(x-1)(x+2)^2} \, dx$
(ii) $\int \frac{1}{x^3-1} \, dx$

Hamdard University Bangladesh

Dept. of Electrical & Electronic Engineering

Final Examination (Fall'24)

PHY111 Physics I

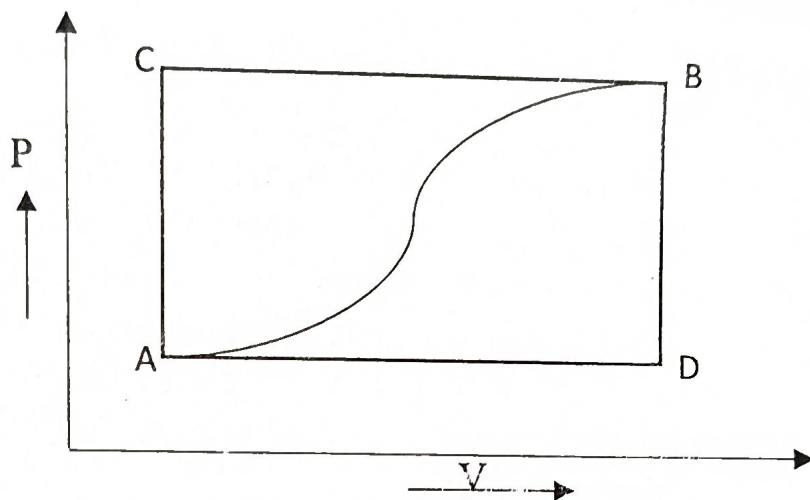
Time: 2.00 hours

Marks: 40

Answer any 4 (Four) of the following sets of questions

- 1) (a) From the interference due to transmitted light on thin films, find the equation of path difference for bright fringes and dark fringes respectively. 6 CLO2
(b) In an experiment with Fresnel's biprism, fringes for light of wavelength 5×10^{-5} cm are observed 0.2 mm apart at a distance of 175 cm from the prism. The prism is made of glass of refractive index 1.50 and it is at a distance of 25 cm from the illuminated slit. Calculate the angle at the vertex of the biprism. 4 CLO3
- 2) (a) In a Newton's rings experiment the diameter of the 12th ring changes from 1.50 cm to 1.35 cm when a liquid is introduced between the lens and the plate. Calculate the refractive index of the liquid. 3 CLO3
(b) In a Fraunhofer diffraction due to a narrow slit, a screen is placed 2 m away from the lens to obtain the pattern. If the slit width is 0.2 mm and the first minima lies 5 mm on either side of the central maximum, find the wavelength of light. 3 CLO3
(c) From the Brewster's Law of polarization, show that the reflected and the refracted rays are at right angles to each other. Also find the polarization angle for refractive index of 1.52 from the Brewster's Law. 4 CLO2
- 3) (a) From the Fraunhofer diffraction at a circular aperture, show that $x = \frac{f\lambda}{a}$ and also sketch the intensity distribution on the screen. 6 CLO2
(b) A thin equiconvex lens of focal length 4 meters and refractive index of 1.50 rests on and in contact with an optical flat, and using light of wavelength 5460 Å. Newton's ring are viewed normally by reflection. What is the diameter of the 5th bright ring? 4 CLO3
- 4) (a) Write down the postulates of the kinetic theory of gases. 3 CLO1
(b) Show that the energy associated with each degree of freedom is $\frac{1}{2} kT$. 4 CLO2
(c) At what temperature, pressure remaining constant, will the R.M.S velocity of a gas be half its value at 0° C. 3 CLO3

- (a) Show that the R.M.S velocity of a gas $C = \sqrt{\frac{3P}{\rho}}$. 6 CLO2
- (b) When a system is taken from the state A to the state B, along the path ACB, 80 J of heat flows into the system, and the system does 30 J of work. 4 CLO3
- (i) How much heat flows into the system along the path ADB, if the work done is 10 J.
- (ii) The system is returned from the state B to the state A along the curved path. The work done on the system is 20 J. Does the system absorb or liberate heat and how much?
- (iii) If $U_A = 0$, $U_D = 40$ J, find the heat absorbed in the process AD and DB



- (a) State the zeroth law of thermodynamics. 2 CLO1
- (b) Show that for an adiabatic reversible process, $\left(\frac{\delta T}{\delta V}\right) = \frac{C_V - C_P}{\alpha V C_V}$, where α is the isobaric volume coefficient of expansion. 4 CLO2
- (c) Calculate the mean free path of a gas molecules in a chamber of 10^{-6} mm of mercury pressure, assuming the molecular diameter to be $2 \text{ \AA}$. One gram molecule of the gas occupies 22.4 litres at N.T.P. Take the temperature of the chamber to be 273 K. 4 CLO3

$\frac{C_P - C_V}{\alpha V C_V}$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering

ENG111 English Basics

Final Examination

Fall 2024

Batch 27th

March 19, 2025

1st Semester

Time: 2 Hours

Total Marks: 40

No.	Question/ Statement	CLOs	Level	Marks
1.	Read the following passage and fill in the blanks with the correct form of the verbs in brackets. Education (a) _____ (be) the backbone of a nation. No nation can (b) _____ (prosper) without education. It (c) _____ (help) us to distinguish between right and wrong. An educated person (d) _____ (respect) in society. If people (e) _____ (educate), they (f) _____ (know) how to lead their lives properly. Unfortunately, many people in our country (g) _____ (not realize) the importance of education. They (h) _____ (think) it is a waste of time. The government (i) _____ (take) various measures to spread education. We all (j) _____ (work) together to build an educational nation.	CLO5	Apply	10 × 0.5 = 5
2.	Read the following passage and identify the nouns. In a small town, there was a famous doctor named Rahim. He treated his patients with great care. People from different places came to visit him. His kindness and honesty made him a respected figure in society.	CLO5	Apply	10 × 0.5 = 5
3.	Read the following passage and fill in the blanks with appropriate articles (a, an, the) or 'x' where no article is needed. Aristotle was (a) _____ ancient Greek philosopher and scientist. He was born in Stagira, (b) _____ small town in Greece. He was (c) _____ student of Plato and later became (d) _____ teacher of Alexander the Great. Aristotle made great contributions to (e) _____ field of philosophy, science, and logic. He believed that (f) _____ purpose of life was to achieve happiness through virtue and wisdom. His writings covered various subjects, including politics, ethics, and biology. Even today, he is considered (g) _____ one of the greatest thinkers in history. Many of his ideas influenced (h) _____ modern scientific and philosophical thought. Aristotle founded (i) _____ school called the Lyceum, where he taught his students. His works continue to inspire scholars around (j) _____ world.	CLO5	Apply	10 × 0.5 = 5

4. Read the passage below and fill in the blanks with appropriate prepositions.

CLO5 Apply $10 \times 0.5 = 5$

Rahim is a student (a) _____ Class 10. He is good (b) _____ mathematics and always pays attention (c) _____ his studies. He is keen (d) _____ learning new things. Every day, he goes (e) _____ school early and sits (f) _____ the front row. His teachers are pleased (g) _____ him because he is hardworking. He never gives up (h) _____ any challenge. His parents are proud (i) _____ him and always encourage him to do better. He dreams (j) _____ becoming a doctor in the future.

5. Write a letter to the Manager of a Local Electricity Office complaining about frequent power cuts in your area, explaining how it is affecting the daily lives of students and residents. Request an immediate solution to the issue.

CLO4 Apply $08 \times 1 = 08$

6. Write a paragraph from the following alternatives.

- a. Importance of Reading Books
- b. The Benefits of Exercise
- c. Crime in Bangladesh.

CLO4 Apply $12 \times 1 = 12$

What is crime
Result in crime
Why crime happens

The

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 117: Engineering Ethics and Cyber Law

Batch: 27th

Date: 15/03/2025

Final Examination

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

1. ~~(a)~~ What is ransomware? How does it differ from phishing attacks? CLO5 2
~~(b)~~ Define cybercrime and explain two main types with examples. CLO5 3
~~(c)~~ Demonstrate the key challenges in establishing jurisdiction in cyberspace. CLO4 3
2. A company experiences a data breach because an employee disabled the firewall to speed up access to certain websites. As a result, customer data was leaked online
 - (a) Suggest two policies the company could implement to prevent such incidents in the future. CLO5 2
 - (b) What legal responsibilities does the company have regarding customer data protection? CLO5 3
 - (c) What role does a firewall play in network security, and how could it have prevented this breach? CLO5 3
3. ~~(a)~~ Relate between Proper Engineering Decisions (PED) and Proper Management Decisions (PMD) CLO2 4
~~(b)~~ What is whistle-blowing? Explain the conditions under which whistle-blowing becomes morally obligatory. CLO2 4
4. A popular social media influencer in Bangladesh has approached your software company with the following requests:
 - To create multiple fake accounts using real people's photos and information.
 - To develop a bot that automatically downloads and reposts viral content from other creators.
 - To implement a system that artificially inflates view counts and engagement metrics.
 - ~~(a)~~ Identify which specific provisions of the Digital Security Act 2018 and ICT Act 2006 would be violated by these requests CLO4 4
 - (b) As the lead engineer assigned to this project:
 - What ethical considerations should you weigh? CLO2 4
 - How would you handle this situation while maintaining professional integrity?

5. A university student gains unauthorized access to their academic grading system and changes their grades. The student later argues that they did not intend to cause harm and were only testing system security.

(a) Is this act considered ethical under professional computing standards? CLO2 4
Defend your answer

(b) How did the meaning and implications of the term "Hacking" evolve from the 1960s to the Web era? CLO4 4

6. (a) Define Intellectual Property (IP). What are its key types? CLO3 2

(b) Explain the differences between copyright, patent, trademark, and trade secret. CLO3 3

(c) How does the Bangladesh Copyright Act, 2000 protect the rights of content creators? CLO4 3

7. (a) Determine the tension between crime fighting and civil liberties in cybersecurity. What are the key considerations for balancing between crime fighting and civil liberties? CLO4 4

(b) Explain how Network Address Translation (NAT) contributes to network security? Mention one limitation of firewalls in providing complete network protection. CLO5 4